

1) Classification Algorithm

a) Age<30 & Govt satisfaction less, Urban

Trained Model Score :

THE TRAINED MODEL

Party A score = **-11.7** pts ← base points
(starting score) + **0.012**×Age + **2.20**×Govt.
satisfaction - **0.795**×Urban voter

Party T score = **11.1** pts ← base points
(starting score) - **0.138**×Age - **1.34**×Govt.
satisfaction + **1.36**×Urban voter

Party C score = **0.387** pts ← base points
(starting score) + **0.089**×Age - **0.748**×Govt.
satisfaction - **1.07**×Urban voter

Undecided score = **0.904** pts ← base points
(starting score) - **0.036**×Age + **0.513**×Govt.
satisfaction - **0.123**×Urban voter

Winner Party Name after a Prediction based on above Criteria age<30 & Govt Less satisfaction & Urban Selected

MAKE A PREDICTION

Age 30.0 yrs

Govt. satisfaction 1.80 /10

Urban voter Yes

Predict

PREDICTED PREFERENCE **Party T**

Party A
0.0%

Party T
99.0%

Party C
0.5%

Undecided
0.5%

Step 1 — each class gets its own score = base + (weight × this voter's value), added up **Party A**

$$\text{-11.7 pts} \leftarrow \text{base points} + 0.012 \times 30.0 \text{ (Age)} = +0.370 + 2.20 \times 1.80 \text{ (Govt. satisfaction)} = +3.96 - 0.795 \times \text{Yes (Urban voter)} = -0.795 \text{ —————} = \text{-8.12 pts}$$

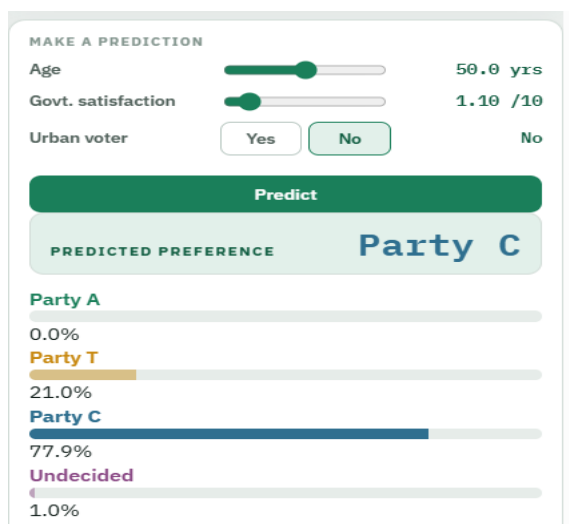
$$\text{Party T 11.1 pts} \leftarrow \text{base points} - 0.138 \times 30.0 \text{ (Age)} = -4.15 - 1.34 \times 1.80 \text{ (Govt. satisfaction)} = -2.40 + 1.36 \times \text{Yes (Urban voter)} = +1.36 \text{ —————} = \text{5.88 pts}$$

$$\text{Party C 0.387 pts} \leftarrow \text{base points} + 0.089 \times 30.0 \text{ (Age)} = +2.66 - 0.748 \times 1.80 \text{ (Govt. satisfaction)} = -1.35 - 1.07 \times \text{Yes (Urban voter)} = -1.07 \text{ —————} = \text{0.629 pts}$$

$$\text{Undecided 0.904 pts} \leftarrow \text{base points} - 0.036 \times 30.0 \text{ (Age)} = -1.07 + 0.513 \times 1.80 \text{ (Govt. satisfaction)} = +0.923 - 0.123 \times \text{Yes (Urban voter)} = -0.123 \text{ —————} = \text{0.638 pts}$$

————— highest score wins → **Party T**

b) Age >50 and Govt satisfaction less ,non urban validate the percentage



Step 1 — each class gets its own score = base + (weight × this voter's value), added up **Party A**

$$\text{-12.6 pts} \leftarrow \text{base points} + 0.040 \times 50.0 \text{ (Age)} = +1.99 + 2.13 \times 1.10 \text{ (Govt. satisfaction)} = +2.34 - 0.757 \times \text{No (Urban voter)} = +0 \text{ —————} = \text{-8.27 pts}$$

$$\text{Party T 10.1 pts} \leftarrow \text{base points} - 0.111 \times 50.0 \text{ (Age)} = -5.55 - 1.41 \times 1.10 \text{ (Govt. satisfaction)} = -1.55 + 1.40 \times \text{No (Urban voter)} = +0 \text{ —————} = \text{3.04 pts}$$

$$\text{Party C -0.562 pts} \leftarrow \text{base points} + 0.116 \times 50.0 \text{ (Age)} = +5.81 - 0.820 \times 1.10 \text{ (Govt. satisfaction)} = -0.902 - 1.03 \times \text{No (Urban voter)} = +0 \text{ —————} = \text{4.35 pts}$$

$$\text{Undecided -0.046 pts} \leftarrow \text{base points} - 0.008 \times 50.0 \text{ (Age)} = -0.401 + 0.441 \times 1.10 \text{ (Govt. satisfaction)} = +0.485 - 0.084 \times \text{No (Urban voter)} = +0 \text{ —————} = \text{0.038 pts}$$

————— highest score wins → **Party C**

2) Linear Regression:

a) QA Salary:

Trained Model Calculation:

THE TRAINED MODEL

```
Salary_LPA in LPA =  
3.10 LPA + base salary (minimum)  
+ 0.966 LPA per year × Experience  
+ 1.34 LPA per point × Skill rating
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Prediction made for First Exp>12 & Skill<5.5

MAKE A PREDICTION

Experience 12.0 yrs
Skill rating 5.30 /10

Predict

TOTAL PACKAGE 21.8 LPA

Base salary – the starting value, before any feature counts
3.10 LPA

Experience
0.966 LPA per year × 12.0 yrs
= +11.6 LPA

Skill rating
1.34 LPA per point × 5.30 /10
= +7.11 LPA

TOTAL 21.8 LPA

Prediction made for Exp>12 & Skill >7.5

MAKE A PREDICTION

Experience 12.0 yrs
Skill rating 7.60 /10

Predict

TOTAL PACKAGE 24.9 LPA

Base salary – the starting value, before any feature counts
3.10 LPA

Experience
0.966 LPA per year × 12.0 yrs
= +11.6 LPA

Skill rating
1.34 LPA per point × 7.60 /10
= +10.2 LPA

TOTAL 24.9 LPA

b) Food Delivery Time

Trained Model Calculation:

THE TRAINED MODEL

Delivery_Minutes in min =

$$\begin{aligned} & 7.56 \text{ min} + \text{base time (minimum)} \\ & + 2.40 \text{ min per km} \times \text{Distance} \\ & + 3.16 \text{ min per point} \times \text{Traffic} \end{aligned}$$

First Prediction Distance > 15 & Traffic < 4

MAKE A PREDICTION

Distance 15.0 km

Traffic 4.00 /10

Predict

TOTAL TIME 56.2 min

Base time – the starting value, before any feature counts
7.56 min

Distance
 $2.40 \text{ min per km} \times 15.0 \text{ km}$
= +36.0 min

Traffic
 $3.16 \text{ min per point} \times 4.00 /10$
= +12.6 min

TOTAL 56.2 min

Second Prediction Distance > 15 & Traffic > 6

MAKE A PREDICTION

Distance 15.0 km

Traffic 6.10 /10

Predict

TOTAL TIME 62.9 min

Base time – the starting value, before any feature counts
7.56 min

Distance
 $2.40 \text{ min per km} \times 15.0 \text{ km}$
= +36.0 min

Traffic
 $3.16 \text{ min per point} \times 6.10 /10$
= +19.3 min

TOTAL 62.9 min